Topics/Subjects

Provide a list of topics, themes, or questions for students to choose and complete according to the requirements.

Topic 1: Student Learning Habits Data Collection and Cleaning

- **Problem Description:** Collect and preprocess data about university students' learning habits and study performance.
- **Suggested Libraries:** numpy, pandas, matplotlib, openpyxl
- **Detailed Requirements:**
  - Design an online questionnaire.
  - Collect at least 500 responses.
  - Suggested attributes:
    - Year of study
    - Study time
    - Self-study time
    - Sleep duration
    - Social media usage
    - Attendance frequency
    - Academic performance
  - Detect missing, duplicate, invalid, inconsistent, and outlier values.
  - Standardize categorical and numerical values.
  - Perform suitable data transformation.
  - Compare data quality before and after cleaning.
  - Export Raw Data and Cleaned Data separately.
- **Suggested Data Sources:**
  - Students from the university
  - Students from the same faculty/program
  - Google Forms
  - Microsoft Forms
  - Direct interviews
- **Data Collection Requirement:** Students must design and conduct the survey themselves. Existing datasets are not allowed.

Topic 2: Consumer Shopping Behavior Data Collection and Cleaning

- **Problem Description:** Collect consumer shopping behavior data and prepare a consistent dataset for further analysis.

- **Suggested Libraries:** numpy, pandas, matplotlib, openpyxl
- **Detailed Requirements:**
  - Collect at least 1000 consumer responses.
  - Suggested attributes:
    - Age group
    - Occupation
    - Shopping frequency
    - Online/offline preference
    - Monthly spending
    - Product category
    - Payment method
    - Promotion usage
    - Satisfaction
  - Detect missing and duplicated records.
  - Standardize payment methods and product categories.
  - Detect unusual spending values.
  - Create at least 2 derived attributes.
  - Export cleaned data.
- **Suggested Data Sources:**
  - University students
  - Family members and acquaintances
  - Customers at supermarkets or convenience stores
  - Online shopping communities
  - Google Forms / Microsoft Forms
  - Direct interviews
- **Data Collection Requirement:** Each group must collect its own responses. Kaggle/UCI datasets are not accepted.

Topic 3: Restaurant Menu Data Collection and Cleaning

- **Problem Description:** Collect menu information from multiple restaurant websites and integrate it into a clean dataset.
- **Suggested Libraries:** requests, beautifulsoup4, selenium, pandas, re
- **Detailed Requirements:**
  - Collect at least 300 menu items.
  - Collect:
    - Restaurant
    - Product name

        - Category
        - Price
        - Description
        - Promotion
    - Remove HTML and special characters.
    - Normalize price formats.
    - Handle missing categories or descriptions.
    - Remove duplicates.
    - Integrate information from multiple sources.
- **Suggested Data Sources:**
    - Official restaurant websites
    - Restaurant online menus
    - Public food-ordering pages where collection is permitted
    - Restaurant Facebook pages with publicly displayed menus
    - Restaurant promotional pages
- Possible categories:
    - Coffee shops
    - Fast food
    - Pizza
    - Milk tea
    - Vietnamese restaurants
- **Data Collection Requirement:** Students must write their own collection program or manually collect information from the selected public sources. Existing menu datasets are not allowed.

Topic 4: Online Product Price Data Collection and Cleaning

- **Problem Description:** Collect product information from multiple online retailers and construct a standardized product comparison dataset.
- **Suggested Libraries:** requests, beautifulsoup4, selenium, pandas, numpy, re
- **Detailed Requirements:**
    - Choose one product category.
    - Collect at least 1000 products from at least 10 sources.
    - Collect:
        - Product name
        - Brand
        - Current price
        - Original price

        - Rating
        - Seller/store
        - Product URL
    - Normalize product names.
    - Standardize monetary values.
    - Handle missing ratings/prices.
    - Detect duplicated products across different stores.
    - Combine all sources into one dataset.
- **Suggested Data Sources:**
    - Thegioididong
    - FPT Shop
    - CellphoneS
    - Phong Vũ
    - Nguyễn Kim
    - Điện Máy Xanh
    - Fahasa
    - Tiki public product pages where permitted
- Possible product categories:
    - Smartphones
    - Laptops
    - Monitors
    - Headphones
    - Keyboards
    - Books
    - Household appliances
- **Data Collection Requirement:** Students must collect current product information directly from selected websites. Downloading an existing price dataset is not accepted.

Topic 5: News Headlines Collection and Text Cleaning

- **Problem Description:** Collect news headlines from multiple online sources and build a clean text dataset.
- **Suggested Libraries:** requests, beautifulsoup4, pandas, nltk, re
- **Detailed Requirements:**
    - Collect at least 500 news headlines from at least 5 sources.
    - Collect:
        - Headline

        - Category
        - Date/time
        - News source
        - URL
    - Remove duplicates.
    - Standardize publication dates.
    - Clean unnecessary characters.
    - Convert text to lowercase.
    - Perform tokenization and stopword removal.
    - Compare raw text with cleaned text.
- ❖ **Suggested Data Sources:**
    - VnExpress
    - Tuổi Trẻ
    - Thanh Niên
    - VietnamNet
    - Dân Trí
    - Lao Động
    - Official RSS feeds if available
    - Public news webpages where collection is permitted
- ❖ **Data Collection Requirement:** The news records must be collected by students during the project period. Existing news datasets are not allowed.

Topic 6: Real-Time Weather Data Collection and Cleaning

- ❖ **Problem Description:** Collect weather observations through a web API and construct a time-series dataset.
- ❖ **Suggested Libraries:** vrequests, json, pandas, datetime, matplotlib
- ❖ **Detailed Requirements:**
    - Select at least 6 cities or locations.
    - Collect at least 600 observations.
    - Collect:
        - Time
        - Temperature
        - Humidity
        - Wind speed
        - Precipitation
        - Pressure
    - Parse JSON responses.

- Standardize timestamps.
- Detect missing API responses.
- Detect duplicated observations.
- Validate measurement ranges.
- Combine observations over time.

❖ **Suggested Data Sources:**

- Open-Meteo API
- OpenWeather API
- WeatherAPI
- Other legitimate public weather APIs

❖ Suggested locations:

- Ho Chi Minh City
- Hanoi
- Da Nang
- Can Tho
- Da Lat
- Phu Quoc

❖ **Data Collection Requirement:** The dataset must be generated by executing the students' own API collection code. Historical datasets downloaded elsewhere are not accepted.

Topic 7: Air Quality Monitoring Data Collection and Cleaning

❖ **Problem Description:** Collect air-quality observations from public monitoring systems or APIs and prepare a clean environmental dataset.

❖ **Suggested Libraries:** requests, json, pandas, numpy, matplotlib

❖ **Detailed Requirements:**

- Select multiple monitoring locations.
- Collect:
  - Time
  - PM2.5
  - PM10
  - CO
  - NO_2
  - SO_2
  - O_3
  - AQI
- Collect data repeatedly or across multiple locations.

- Handle missing sensor measurements.
- Detect invalid or impossible readings.
- Standardize measurement units.
- Merge monitoring data by location and time.

❖ **Suggested Data Sources:**

- OpenAQ
- IQAir public information
- Government environmental monitoring portals
- Public environmental APIs
- Local monitoring station webpages

❖ **Data Collection Requirement:** Students must retrieve the observations themselves through API requests or permitted public sources. Ready-made air-quality datasets are not allowed.

Topic 8: Job Posting Data Collection and Cleaning

❖ **Problem Description:** Collect online job advertisements and create a clean dataset describing current recruitment demand.

❖ **Suggested Libraries:** requests, beautifulsoup4, selenium, pandas, re

❖ **Detailed Requirements:**

- Select one job field, for example:
  - Data Analyst
  - Software Developer
  - Marketing
  - Accounting
  - Sales
- Collect at least 1000 job postings.
- Collect:
  - Job title
  - Company
  - Location
  - Salary
  - Experience requirement
  - Skills
  - Posting date
  - URL
- Normalize job titles.
- Standardize salary formats.

- Clean skill descriptions.
- Handle missing salary values.
- Remove duplicate job postings.

❖ **Suggested Data Sources:**

- TopCV
- VietnamWorks
- CareerViet
- ITviec
- LinkedIn public job pages where access is permitted
- Company career websites

❖ **Data Collection Requirement:** Students must collect job postings during the project period. Previously prepared recruitment datasets are not accepted.

Topic 9: University Course Information Collection and Integration

❖ **Problem Description:** Collect course information from university websites and integrate data into one standardized dataset.

❖ **Suggested Libraries:** requests, beautifulsoup4, pandas, re, openpyxl

❖ **Detailed Requirements:**

- Collect at least 500 course records.
- Suggested attributes:
  - Course code
  - Course name
  - Credits
  - Faculty
  - Semester
  - Prerequisite
  - Course description
- Collect data from multiple pages or faculties.
- Standardize course codes and names.
- Detect duplicate courses.
- Handle missing descriptions/prerequisites.
- Join course data with faculty/program information.

❖ **Suggested Data Sources:**

- University academic webpages
- Faculty webpages
- Public curriculum pages
- Public course catalogs

- Official program descriptions

❖ **Data Collection Requirement:** Students must build the dataset from public university information themselves.

Topic 10: Public Transportation Information Collection and Cleaning

❖ **Problem Description:** Collect public transportation information and construct a standardized route/station dataset.

❖ **Suggested Libraries:** requests, json, pandas, beautifulsoup4, geopandas (optional)

❖ **Detailed Requirements:**

- Collect information about at least 200 routes or 400 stops.
- Suggested attributes:
  - Route number
  - Origin
  - Destination
  - Stop name
  - Operating time
  - Ticket price
  - Location
  - Frequency
- Standardize station and route names.
- Handle missing schedules.
- Detect duplicated stops.
- Normalize time and price formats.
- Join route information with stop information.

❖ **Suggested Data Sources:**

- Public bus websites
- City transport department webpages
- GTFS feeds if publicly available
- Metro/bus operator webpages
- Open public transport APIs

❖ **Data Collection Requirement:** Students must retrieve current transport information themselves. Existing transport datasets may be used only as a technical reference, not as the project dataset.

The groups can follow this suggested outline:

+ Title page (university name, faculty, course, topic title, group member names & student IDs)

+ Abstract / Introduction: Briefly describe the purpose of the project, the data collection problem, selected data sources, and expected outcomes.
+ Chapter 1: Data Collection and Dataset Description

  Describe the data collection method, data sources, collected attributes, data structure, number of records, and raw data characteristics.
+ Chapter 2: Data Cleaning and Preprocessing

  Present the methods used to identify and handle missing values, duplicate records, invalid data, inconsistent formats, outliers, data transformation, normalization/standardization, restructuring, and data integration if applicable.
+ Chapter 3: Results and Data Quality Evaluation

  Compare the dataset before and after cleaning; report changes in missing values, duplicates, invalid records, consistency, number of records, and other relevant data-quality indicators. Present the final cleaned dataset and selected visualizations or descriptive statistics.
+ Conclusion and Future Work

  Summarize the data collection and cleaning process, major problems identified, improvements achieved, limitations, and possible future extensions.
+ References

  List websites, APIs, documents, tools, libraries, and other sources used during data collection and preprocessing.